

Java Array Practice Program

Note: The original code contained a few Java syntax/logic issues (println without System.out, missing semicolons, and an invalid attempt to decrease array.length). Those parts were corrected in this PDF.

```
public class Main
{
    public static void main(String[] args) {

        // int[] array; declaration
        int[] array = {1, 2, 3, 4, 5}; // initialization

        System.out.println(array); // prints array reference/address

        // Traversal
        for (int i = 0; i < array.length; i++) {
            System.out.println("Array value: " + array[i]);
        }

        // Sum of array
        int sum = 0;
        for (int i = 0; i < array.length; i++) {
            sum = sum + array[i];
        }
        System.out.println("Sum: " + sum);

        // Subtraction of array elements
        int sub = 0;
        for (int i = 0; i < array.length; i++) {
            sub = sub - array[i];
        }
        System.out.println("Subtraction: " + sub);

        // Multiplication of array elements
        int mul = 1;
        for (int i = 0; i < array.length; i++) {
            mul = mul * array[i];
        }
        System.out.println("Multiplication: " + mul);

        // Even numbers
        for (int i = 0; i < array.length; i++) {
            if (array[i] % 2 == 0) {
                System.out.println("Even: " + array[i]);
            }
        }

        // Odd numbers
        for (int i = 0; i < array.length; i++) {
            if (array[i] % 2 != 0) {
                System.out.println("Odd: " + array[i]);
            }
        }
    }
}
```

Expected results: Traversal → 1, 2, 3, 4, 5 Sum → 15 Subtraction → -15 Multiplication → 120 Even → 2, 4 Odd → 1, 3, 5